

Dmitry Gladkikh

C++ / Python Developer — Computer Vision, Measurement Systems, Backend

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Website: gladkih.xyz | Novosibirsk | Remote, hybrid, open to relocation

Summary

Developer experienced in building computer vision systems, measurement software, and backend services. In C++ - high-performance computational libraries and CV algorithms; in Python - application systems, data processing, measurement algorithms, GUI, and REST API. Designed an extensible camera-control platform that several measurement systems are built on. I run my own infrastructure: self-hosted GitLab, CI/CD, production services.

Skills

Languages: Python, C++

Computer Vision: OpenCV, OpenCL, AruCo, camera calibration

Backend: Django, DRF, FastAPI, PostgreSQL, Redis, Celery

Desktop / GUI: PySide6, multiprocessing

Infrastructure: Linux, Docker, nginx, GitLab CI/CD, WireGuard, Tailscale, Let's Encrypt

Tools: Git, pytest, Sphinx

Experience

Aviaspetstest - Software Engineer (C++, Python)

December 2023 - present, Novosibirsk

- **Industrial camera-control platform.** Designed and implemented an extensible data-acquisition platform: a proxy layer for integrating different camera SDKs behind a unified interface, synchronized capture from multiple cameras on separate processes (bypassing the GIL), and a core independent of the access method (GUI / REST API). Became the foundation for the other measurement systems.
- **Vibration analysis system.** Contactless measurement of oscillation amplitude and frequency from video (replacing accelerometers): 80 measurements/sec per marker, up to 7 markers. AruCo marker detection, parameter computation, communication with a single-board computer over UDP.
- **Deformation measurement system (stereo-DIC).** C++ CV libraries: target-template recognition, camera calibration, a custom point-matching algorithm across frames. Worked with 25 MP frames.
- **Drawing-digitization scanner and angle-measurement system** - applied systems built on the platform (OpenCV, SQLAlchemy).

Personal Projects

- **CRM for a family business (in progress).** Django + PostgreSQL, in production on a VPS with its own CI/CD via a self-hosted GitLab Runner.
- **Home infrastructure.** Self-hosted GitLab CE, CI/CD pipelines, deployment to production servers over SSH, a private network over Tailscale, automated backups of configs and databases.
- **Portfolio site (gladkih.xyz).** Astro, deployed with nginx SNI routing, HTTPS via Let's Encrypt, automated deployment through GitLab CI/CD.

Education

SSUGT - BSc in Geodesy and Remote Sensing, 2023

Courses: Python Developer (Skypro, 2024) - Django, DRF, PostgreSQL, Celery, Redis; C++ Developer (GeekBrains, 2022)

Additional

English: B2. **Work format:** remote, hybrid, open to relocation.